GERGELY PAPP

DATA SCIENTIST & CONSULTANT

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SKILLS

Deep Learning | LangChain | PostgreSQL | TensorFlow | PyTorch | Python | Docker | Cloud | LLMs | Research | AWS | S3 | Bedrock

Adaptibility | Initiative | Problem Solving | Teamwork | Communication | Teaching

LANGUAGES

- Hungarian ●●●●●
- English ●●●●○
- French ●○○○○

HOBBIES

Gym | Board games | Piano | Pool billiard | Video games

EDUCATION

Master of Artificial Intelligence
University of Amsterdam
Grade: 8.1 (Cum Laude) *2024*

Bachelor of Computer Science
University of Manchester
Grade: Second-Upper Class *2017*

WORK EXPERIENCE

Data Scientist & Consultant 01/2026 - present
Datacation B.V.

- Delivering projects end-to-end, from quick scans to production
- Advising on traditional ML methods when more suitable than LLMs
- *Azure, Huggingface, Computer Vision, Agentic AI*

AI Engineer 06/2025 - 01/2026
GergoTech - Freelancer

- **Built a RAG, processing 1000+ page files for US law enforcement**
- Design and deploy RAG systems with knowledge graphs
- Implement scalable LLM agentic pipelines
- *LangChain, Bedrock, Docker, PostgreSQL, S3, and Redis*

Machine Learning Consultant 10/2020 - 07/2024
Asura Technologies Ltd. *(part time)*

- **Boosted object detection accuracy from 75% to 93%**
- Guided a group of ML developers on computer vision projects
- Advised on technology & implementations to streamline workflows

Machine Learning Engineer
Asura Technologies Ltd. 10/2018 - 10/2020

- **Drove startup growth from 12 to 100+ members**
- **Engineered an in-house license plate recognition app from scratch that outperformed competitors both in speed and accuracy**
- Design, train and serve real-time **object detection** models
- Create and maintain an **ALPR** and **OCR** engine, as well as an automated parking system that tracks cars in a parking lot
- Deliver state-of-the-art PoC models for new customers
- *TensorFlow, Scikit-Learn, OpenCV, Django, Flask, C#, Docker, Git*

Deep Learning Research Engineer *full-time 10/20 - 09/22*
Alfréd Rényi Institute of Mathematics *part-time 09/22 - 07/24*

- **Published the first NeurIPS paper from a Hungarian institute**
- Literature review and writing **conference papers**
- Build scalable training pipelines for model training and evaluation
- *PyTorch, Pandas, GAN, VAE, Transformers, Huggingface*

EXT. STUDIES

Coursera courses

- Mathematics for Machine Learning: Linear Algebra
- Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization
- Structuring Machine Learning Projects
- Neural Networks and Deep Learning

Summer schools

2018 - Cluj - DeepMind - Machine Learning Summer School

PUBLICATIONS

2023 - **Neural Networks** - Mode combinability: Exploring convex combinations of permutation aligned models

2023 - **ReScience** - Reproducibility study of "Label-Free Explainability for Unsupervised Models"

2018 - **AITP** - Ordering Subgoals in a Backward Chaining Prover

2018 - **NeurIPS** - Similarity and Matching of Neural Network Representations

HACKATHONS

2015 MLH - Manchester, UK
2016 MLH - Sheffield, UK
2016 Ultrahack - Helsinki, FI
2018 HackPrague - Prague, CZ
2019 LikeABosch - Budapest, HU

NOSTALGIC

Pool billiard
European Champion, 2010

High School
Fazekas Mihaly High School
Specialized in Mathematics

National Secondary School Academic Competition
27. place in Mathematics
24. place in Programming

University of Amsterdam — 10/2024 - 12/2024

Teaching Assistant (Part Time)

- **Successfully helped students to understand the fundamentals of deep learning and different neural architectures**
- Teach and design curriculum for AI master students.
- Enhance course quality through program evaluation
- Mark students by carefully designed unit tests
- Hold tutorials and Q&A sessions for students
- *Python, SLURM, PyTorch, NumPy*

Morgan Stanley — 08/2017 - 05/2018

Risk Analyst (AI team)

- **Automated data processing workflows, saving 100hrs / week.**
- Save working hours by automating quick-decision processes
- Develop clustering and forecasting models on tabular data
- Learn about the banking industry while being an expert of coding
- Train light-weight traditional ML algorithms on big data
- *Python, SKlearn, Pandas, Spark, SQL, Q, Excel*

PROJECTS

- **WRPGMaps (2026):** Built a generative workflow to generate virtual tabletop D&D maps anchored to AI generated locations, given a theme. Automatically corrected visual AI artifacts. Generated a brief storyline and added several NPCs with detailed descriptions and generated their visual appearances. [Python, Google Cloud, Claude Code, Codex, Cursor]
- **Linear D&D (2025):** Generated a linear D&D storyline using a game tree, local LLMs, and APIs, supported by a set of story-writing and D&D expert agents. Ensured that rules and statistics aligned with the storyline and that the story was enjoyable and playable. [Python, OpenRouter, Ollama, Claude Code]
- **TruthWorks (2025):** AI-powered investigation platform for US police departments featuring OCR, image captioning, and RAG system processing thousands of pages to identify contradictions in case files. [AWS Bedrock]
- **Prisma (2025):** Multi-LLM framework for influencers to schedule and automate Instagram posts, featuring full-stack development from backend to cloud deployment on AWS. [AWS Bedrock, LangChain, Strands]
- **OSChat (2025):** A terminal app that distinguishes English prompts from bash prompts, working like Claude Code. [PyTorch, Docker, JavaScript, Flask]
- **Stitch-BERT (2024):** Analyzed how NLP transformers fine-tuned for different languages and tasks relate geometrically and functionally, revealing potential for cross-task insights. [PyTorch, Python]
- **RAG (2024)**: A project involving vector databases, knowledge graphs and text generation with LLMs for answering questions on many documents.
- **Gaming Bot (2024):** Developed a rule-based AI in NodeJS for automating gameplay for a browser game. The bot timed attacks, reacted to attack reports, and logged summaries to a dashboard, saving significant time.
- **MSc Thesis (2023):** Investigated Vision Transformers' ability to generalize across object properties (shape, texture, color, count) on CLEVR-4. This project involved the use of vision transformers. [Python, Huggingface]
- **Self-Supervised Learning Toolkit (2022):** Created a pip-package standardizing ImageNet evaluation pipelines for self-supervised learning models, enabling consistent community benchmarking. [PyTorch, Python]
- **Watermeter Reader (2020):** Built an OCR-based Python application to clean, rotate, and detect characters from watermeter images for automated reading. Utilized object detection algorithms. [Python, TensorFlow, Docker]
- **AlphaZero (2018):** Reimplemented AlphaZero to explore temporal difference learning vs. Monte Carlo methods. The study revealed unique in-game strategies made with Reinforcement Learning. [Python, Keras]
- **Chess Engine (2017):** Designed a Java-based neural chess engine from scratch without the use of tree search, achieving entry-level play [Java, GraphViz]